

CLASS 9 MATHEMATICS • NCERT 0904CH02

Introduction to Linear Polynomials

An exhaustive point-by-point classroom presentation covering algebraic expressions, univariate polynomials, degree classification, linear patterns, growth & decay models, slope-intercept form, and graph visualization.



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From Physical Realities to Algebraic Expressions

Example 1: Pens & Pencils

- Red boxes have 4 pens each (x boxes = $4x$ pens).
- Blue boxes have 5 pencils each (y boxes = $5y$ pencils).
- Plus 3 extra free pens.
- Total items = $4x + 5y + 3$.

Terms: $4x$, $5y$, 3

Variables: x , y

Coefficients: 4, 5

Constant: 3

Example 2: Garden Cost

- Wire fence along length l @ ₹100/m = $200l$.
- Wooden fence along width w @ ₹80/m = $160w$.
- Seeds sown across area $l \times w$ @ ₹50/m² = $50lw$.
- Total Cost = $200l + 160w + 50lw$.

Contains terms in two variables with product term $50lw$.

Example 3: Bent Wire Area

- Wire of length 20 cm bent into rectangle.
- Semi-perimeter = 10 cm.
- If length = x cm, width = $(10 - x)$ cm.
- Area = $x(10 - x) = 10x - x^2$.

Contains only **one variable (x)** with power 2!

Key Transition: "Letter-Numbers" to "Variables"

The symbols x , y , l , w whose values vary according to the context are formally termed **variables**. When numbers multiply variables, they act as **coefficients**.

Check Your Understanding: Terms, Variables & Constants

Q1 In the expression $4x + 5y + 3$, what are the variables, what are the coefficients, and which term is the constant?

Q2 Why does the garden cost expression $200l + 160w + 50lw$ contain a product term $50lw$, while the pen-pencil expression does not?

Q3 If a 20 cm wire forms a rectangle with length x , why is the width $(10 - x)$ instead of $(20 - x)$?

Q4 What is the fundamental difference between the expressions in Examples 1 & 2 versus the expression in Example 3?

Univariate Polynomials & Their Degree

Definition: Univariate Polynomials & Degree

An algebraic expression involving only **one variable** and whole-number powers of that variable is called a **univariate polynomial**. The **highest power** of the variable is called its **degree**.

0 Degree 0

$$8 = 8x^0$$

Constant Polynomial

Any non-zero constant has degree 0.

1 Degree 1

$$3z + 7$$

Linear Polynomial

Highest power of variable is 1. The focus of this chapter!

2 Degree 2

$$x^2 + 5x + 1$$

Quadratic Polynomial

Highest power of variable is 2 (e.g. $10x - x^2$).

3 Degree 3

$$5y^3 + y^2 - 8$$

Cubic Polynomial

Highest power of variable is 3.

Check Your Understanding: Degree & Coefficients

Q1 What is the degree of each of the following polynomials: (i) $2x^2 - 5x + 3$, (ii) $y^3 + 2y - 1$, (iii) -9 , and (iv) $4z - 3$?

Q2 Why is the constant number -9 regarded as a polynomial of degree 0?

Q3 In the polynomial $x^4 - 3x^3 + 6x^2 - 2x + 7$, what is the coefficient of x^3 and what is the coefficient of x^2 ?

Q4 What is the coefficient of z in the polynomial $4z^3 + 5z^2 - 11$? (Hint: Does a z term appear?)

Linear Polynomials: Constant Rate of Change

Example 4: Perimeter of a Square

- Side of square = $x \Rightarrow$ Perimeter = $4x$.
- Side increases by 0.5 cm (1.0, 1.5, 2.0, 2.5 cm).
- Perimeters: 4, 6, 8, 10 cm.
- **Successive Difference:** Exactly **2 cm** every time!

$$\Delta P = 4 \cdot \Delta x$$

CONSTANT INCREMENT

Example 5: Chess Club Fee

- Joining fee = ₹200; Match fee = ₹50 per match m .
- Total Amount Paid = $200 + 50m$.
- For $m = 1, 2, 3, 4$: Amount = ₹250, ₹300, ₹350, ₹400.
- **Constant Step:** Increases by exactly ₹50 per extra match!

$$\text{Cost} = 200 + 50m$$

LINEAR RATE STRUCTURE

The Definitive Signature of a Linear Polynomial

The difference between successive values of a linear polynomial at consecutive integer values of the variable is ALWAYS **constant**. Such patterns are called **linear patterns**.

Equating to a Constant & The Input–Output Model

= Equating Linear Polynomial = Constant

When we equate a linear polynomial in one variable to a constant, we obtain a **linear equation**.

Example 6: Sum of two numbers is 64; one is 10 more than the other.

- Smaller = x , Larger = $x + 10$
- Linear Equation: $2x + 10 = 64$
- $2x = 54 \Rightarrow x = 27$ (Numbers: 27 and 37)

Polynomials as Functions: Input → Output

Consider $y = 2x + 3$ as a mathematical machine:

- Input $x = 4 \Rightarrow 2(4) + 3 = 11$ (Output $y = 11$)
- Input $x = -6 \Rightarrow 2(-6) + 3 = -9$ (Output $y = -9$)
- Input $x = 0 \Rightarrow 2(0) + 3 = 3$ (Output $y = 3$)

For every input x , there is a unique corresponding output y . Thus, $y = 2x + 3$ is a **linear function** of x !

Think & Reflect: Linear vs. Non-Linear Functions

The expression $2x + 3$ is a *linear function* (degree 1; constant rate of change). In contrast, the rectangular area $10x - x^2$ is a *quadratic function* (degree 2; rate of change varies as x changes).

Check Your Understanding: Equations & Input–Outputs

- Q1** If a player in Example 5 paid ₹750 in total, how many matches did he play under the rule $200 + 50m = 750$?
- Q2** For the linear function $y = 5x - 3$, what are the outputs when the inputs are $x = 0$, $x = -1$, and $x = 2$?
- Q3** Why is the function $y = 2x + 3$ termed 'linear', while $y = 10x - x^2$ is termed 'quadratic'?
- Q4** The sum of Salil's and his mother's ages after 5 years is 70, with mother's present age being 3 times Salil's. What linear equation models this?

Growing Tile Patterns & The General nth Term

Stage (n)	1	2	3	4	5	6	7	General (n)
Number of Tiles	1	3	5	7	9	11	13	$2n - 1$

Mathematical Generalisation

- Every new stage adds **2 square tiles** to the previous one.
- The number of tiles at Stage n is *one less than twice n* .
- General Formula: $T(n) = 2n - 1$.
- Since $T(n)$ has degree 1, this represents a **linear relationship**!

Predictive Power of the nth Term

- **15th Stage:** $2(15) - 1 = 29$ tiles.
- **26th Stage:** $2(26) - 1 = 51$ tiles.
- **Which stage has 47 tiles?**
 $2n - 1 = 47 \Rightarrow 2n = 48 \Rightarrow n = 24$ th Stage.

Linear Sequences in Daily Life

Example 7: Bela's Pocket Money

- Initial pocket money = ₹100; spends ₹5 every day.
- Day 0: ₹100 | Day 1: $100 - 5(1) = ₹95$ | Day 2: ₹90.
- **General Rule on Day n : $A(n) = 100 - 5n$.**
- When does she have ₹40 left?
 $100 - 5n = 40 \Rightarrow 5n = 60 \Rightarrow n = 12\text{th Day}.$
- Balance reaches zero on the **20th Day** ($100 - 5(20) = 0$).

Example 8: Auto-Rickshaw Fare

- Base fare for initial 2 km = ₹25 flat.
- Every kilometre thereafter increases by ₹15/km.
- For travel of 10 km: $25 + 15(10 - 2) = 25 + 120 = ₹145.$
- **General Rule (for $n \geq 2$ km):**
Fare = $25 + 15(n - 2) = 15n - 5.$
- If fare is ₹130: $15n - 5 = 130 \Rightarrow 15n = 135 \Rightarrow n = 9 \text{ km}.$

Formal Definition of a Linear Pattern

A **linear pattern** is a sequence of numbers where the difference between any two consecutive terms is **constant**. The n th term is always expressible as a linear polynomial in n .

Check Your Understanding: Linear Patterns & Rates

- Q1** A student starts with ₹500 in savings and adds ₹150 every month. What is the linear expression for her savings in the n th month?
- Q2** A rally starts with 120 members, and 9 members drop out each hour. How many remain at the end of the n th hour?
- Q3** In Bela's pocket money problem $A(n) = 100 - 5n$, why is the coefficient of n negative (-5)?
- Q4** Sarita is reading a 500-page book at 20 pages per day. What linear expression represents the number of pages left after d days?

Modeling Change: Linear Growth vs. Linear Decay

↗ Linear Growth (Positive Rate)

Definition: Quantity increases by a constant amount over equal intervals.

Example 9: Journey Cost

- Base = ₹100; Rate = ₹60 per km.
- $C(d) = 100 + 60d$.
- For each +1 km, Cost increases by fixed +₹60.
- Travelling 15 km: $100 + 60(15) = ₹1000$.

$$\text{Rate} = +60 > 0$$

CONSTANT UPWARD STEP

↘ Linear Decay (Negative Rate)

Definition: Quantity decreases by a constant amount over equal intervals.

Example 10: Summer Water Tank

- Initial height = 3 m; drops 0.5 m each month t.
- $h(t) = 3 - 0.5t$.
- For each +1 month, height decreases by fixed 0.5 m.
- At end of 5 months: $3 - 0.5(5) = 0.5$ m.

$$\text{Rate} = -0.5 < 0$$

CONSTANT DOWNWARD STEP

Check Your Understanding: Growth vs. Decay

Q1 A plant of initial height 1.75 ft grows 0.5 ft each month. Does this model linear growth or linear decay? Why?

Q2 A phone bought for ₹10,000 depreciates by ₹800 each year. What is the expression for its value $v(t)$ after t years?

Q3 A telecom recharge has ₹600 balance and is reduced by ₹15 each day. After how many days will the balance run out completely?

Q4 What algebraic feature distinguishes a linear growth equation from a linear decay equation?

Formulating Linear Relationships: $y = ax + b$

The General Linear Form

Any linear relationship between two variables x and y can be expressed universally as:

$$y = ax + b$$

STANDARD LINEAR MODEL

- x : independent variable (input)
- y : dependent variable (output)
- a : constant rate of change (slope)
- b : initial fixed value (y -intercept)

Example 11: Mobile Internet Plan

Bill y depends on data used x (in GB) via $y = ax + b$.

- 10 GB cost ₹350 $\Rightarrow 350 = 10a + b$ (Eq 1)
- 20 GB cost ₹550 $\Rightarrow 550 = 20a + b$ (Eq 2)

Solving System:

Subtract Eq 1 from Eq 2: $10a = 200 \Rightarrow a = 20$.

Substitute into Eq 1: $b = 350 - 10(20) = 150$.

Resulting Rule: $y = 20x + 150$ (₹20/GB + ₹150 base fee)

Check Your Understanding: Solving for Parameters a & b

Q1 In Example 11's equation $y = 20x + 150$, what real-world business meanings do the numbers 20 and 150 hold?

Q2 A learning platform charges ₹400 for 10 modules and ₹500 for 14 modules. What is the additional fee per module (a)?

Q3 A gym charges ₹800 for 10 hours and ₹1100 for 15 hours of badminton. What is the fixed monthly fee (b)?

Q4 Given $^{\circ}\text{C} = a(^{\circ}\text{F}) + b$, what equations arise from ice melting ($0^{\circ}\text{C} = 32^{\circ}\text{F}$) and water boiling ($100^{\circ}\text{C} = 212^{\circ}\text{F}$)?

Visualising Linear Equations as Straight Lines

Two Points Fix a Line

To graph any linear equation, we only need to identify **two points**:

For $y = 2x + 1$:

- Set $x = 0 \Rightarrow y = 1 \Rightarrow \mathbf{A(0, 1)}$
- Set $x = 3 \Rightarrow y = 7 \Rightarrow \mathbf{B(3, 7)}$

Plot A and B, join them with a straight ruler, and extend indefinitely in both directions!

Point Satisfaction Rule

If a point lies on the line, its coordinates **must satisfy** the equation.

For $(7, 15)$ on $y = 2x + 1$:

$$\text{LHS} = 15$$

$$\text{RHS} = 2(7) + 1 = 15$$

$$\text{LHS} = \text{RHS} \quad \checkmark$$

Every point on the line satisfies the algebraic equation!

Lines Passing Through Origin

• **Example 12:** Points $(-1, -3)$, $(0, 0)$, $(1, 3)$, $(3, 9)$, $(4, 12)$. y is always 3 times $x \Rightarrow \mathbf{y = 3x}$.

• **Example 13:** Points $(-3, 6)$, $(0, 0)$, $(1, -2)$, $(2, -4)$. y is -2 times $x \Rightarrow \mathbf{y = -2x}$.

Any equation of the form $\mathbf{y = ax}$ passes directly through Origin $(0, 0)$!

Geometric Meanings of Slope (a) & Intercept (b)

The Role of 'a': Slope (Steepness)

- **$a > 0$ (Positive Slope):** Slants upward to the right (models **linear growth**).
- **$a < 0$ (Negative Slope):** Slants downward to the right (models **linear decay**).
- **$|a| > 1$:** Steeper than the line $y = x$ (e.g. $y = 2x$, $y = 3x$).
- **$|a| < 1$:** Less steep / flatter than $y = x$ (e.g. $y = \frac{1}{2}x$, $y = \frac{1}{3}x$).
- **Discrete link:** Slope 'a' represents the constant step difference between consecutive terms!

The Role of 'b': y-Intercept & Parallelism

- Any line $y = ax + b$ crosses the y-axis at **(0, b)**.
- **b is the y-intercept:** Distance from the origin where the line cuts the vertical axis.
- **Fixed a, varying b:** Creates a family of **strictly parallel lines!**
 - $y = 2x - 1$ (cuts at (0, -1))
 - $y = 2x + 1$ (cuts at (0, 1))
 - $y = 2x + 5$ (cuts at (0, 5))
- **Equal Slopes $\Leftrightarrow$ Parallel Lines!**

Check Your Understanding: Graphs, Slopes & Intercepts

Q1 At what point does the straight line $y = 3x - 2$ intersect the vertical y -axis? What is its y -intercept?

Q2 Which line is steeper: $y = 4x$ or $y = 2x$? Which line is flatter: $y = x$ or $y = (1/3)x$?

Q3 Why are the straight lines $y = 5x + 3$ and $y = 5x - 7$ guaranteed to be parallel to each other?

Q4 What is the fundamental visual difference between the graph of $y = 3x + 1$ and the graph of $y = -3x + 1$?

Representative Worked Problems from the Chapter

📌 Celsius & Fahrenheit

Given: $^{\circ}\text{C} = a(^{\circ}\text{F}) + b$. Freezing: (32, 0); Boiling: (212, 100).

Solution:

$$0 = 32a + b \Rightarrow b = -32a$$

$$100 = 212a - 32a = 180a$$

$$\Rightarrow a = 100/180 = 5/9$$

$$\Rightarrow b = -32(5/9) = -160/9$$

$$\left(^{\circ}\text{C} = (5/9)(^{\circ}\text{F} - 32) \right) \checkmark$$

↔ Reversing Digits (Ex 6)

Given: Digits differ by 3; Original + Reversed = 143.

Solution: Let digits be t and u .

$$(10t + u) + (10u + t) = 143$$

$$11(t + u) = 143 \Rightarrow t + u = 13$$

Since $|t - u| = 3$:

- Case 1: $t - u = 3 \Rightarrow 2t = 16 \Rightarrow t = 8, u = 5$ (85)

- Case 2: $u - t = 3 \Rightarrow t = 5, u = 8$ (58)

$$\left(\text{Numbers are 85 and 58} \right) \checkmark$$

/// Slopes & Parallelism (Ex 7)

Identify slopes in $y = ax + b$:

(i) $y = -3x + 4 \Rightarrow \text{slope} = -3$

(ii) $2y = 4x + 7 \Rightarrow y = 2x + 3.5$

(iii) $5y = 6x - 10 \Rightarrow y = 1.2x - 2$

(iv) $3y = 6x - 11 \Rightarrow y = 2x - 11/3$

Lines (ii) & (iv) both have slope $a = 2$. They are parallel! $\checkmark$

Extended Discussions & Real-World Inquiries

Matchstick Hexagons (Problem 12)

A chain of hexagons where each stage adds one hexagon sharing a side:

- Stage 1: 6 matchsticks
- Stage 2: $6 + 5 = 11$ matchsticks
- Stage 3: $11 + 5 = 16$ matchsticks
- **Teacher Prompt:** What is the general rule for Stage n ? (Answer: $5n + 1$). Can a stage ever contain exactly 200 matchsticks? (Hint: Is $(200 - 1)$ divisible by 5?)

Physics: Work Done by Constant Force (Problem 9)

Work done $W = \text{Force } F \times \text{Distance } d$.

- Taking constant force $F = 3$ units: **$W = 3d$** .
- This is a linear equation passing through Origin $(0, 0)$ with slope = 3.
- **Teacher Prompt:** How does plotting this equation on a graph allow you to read off the work done for any fractional distance travelled?

Critical Reflection: Common Property of $f(x) = ax + a$ (Problem 14)

What geometric feature do ALL linear functions of the form $f(x) = ax + a$ ($a > 0$) share in common? (Hint: Evaluate $f(-1) = a(-1) + a = 0$. Every single such line passes through the fixed point $(-1, 0)$ regardless of the value of a !)

Complete Concept Checklist & Core Takeaways

✔ Expressions, Polynomials & Degree

- ✔ **Algebraic Expression:** Combines variables, coefficients, and operations.
- ✔ **Univariate Polynomial:** Expression in one variable with whole-number powers.
- ✔ **Degree:** Highest power of the variable (0 = constant, 1 = linear, 2 = quadratic, 3 = cubic).
- ✔ **Linear Polynomial:** Degree 1 expression of the form $ax + b$ ($a \neq 0$).
- ✔ **Characteristic Signature:** Constant difference between successive integer values.

✔ Patterns, Graphs & Slope-Intercept Form

- ✔ **Linear Growth:** Increases by a constant amount over equal intervals (positive slope).
- ✔ **Linear Decay:** Decreases by a constant amount over equal intervals (negative slope).
- ✔ **Standard Form:** $y = ax + b$, where **a = slope** and **b = y-intercept**.
- ✔ **Origin Line:** When $b = 0$, $y = ax$ passes directly through (0, 0).
- ✔ **Parallel Lines:** Lines having the exact same slope (a) but different intercepts (b).

END OF CHAPTER 2

Lines are the Bridges between Constant Change and Geometry.

"In every steady step, a linear pattern lives; in every straight line, an algebraic truth is revealed."

✦ Akash Srivastva (www.akashmaths.online)